
Causal Variable Audit Debate: Structured Multi-Agent Reasoning for Chinese Commonsense Causal Traps

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Abstract

Large language models often suffer from spurious correlations and *causal parroting*, especially under commonsense causal traps where a single contextual reversal flips the scenario. Multi-agent debate (MAD) is training-free but can *over-correct*: a model that initially answers correctly may be persuaded into an incorrect answer. We propose *Causal Variable Audit Debate* (CVA-Debate), which forces agents to audit core causal variables (X), spurious variables (Z), contextual reversal variables (C), and Minimal Sufficient Evidence (MSE) before rebuttals, with a consistency judge that guards against misclassifying C as Z . We introduce the Over-Correction Rate (OCR) and a small Chinese Causal Trap Diagnostic Set for analysis. Experiments with Qwen2.5-1.5B show that free debate can reach OCR up to 41.9%, while structured auditing reduces OCR by 14.1 percentage points, improving stability while clarifying failure modes such as variable identification errors and hyper-vigilant hallucination of reversals.

1. Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in various reasoning tasks. However, their reliance on statistical patterns in training data often makes them susceptible to spurious correlations, leading to a phenomenon known as “causal parroting.” In natural language scenarios, these models frequently struggle to distinguish between genuine causal relationships and superficial cues that happen to co-occur with the target outcome. This vulnerability is particularly evident in commonsense causal traps, where a single contextual reversal can flip the entire causal trajectory of a scenario.

To address these limitations without the high cost of fine-

tuning, multi-agent debate (MAD) has emerged as a promising, training-free solution. By allowing multiple model instances to propose, critique, and refine answers collectively, MAD aims to leverage the “wisdom of the crowd” to eliminate individual hallucinations and logical errors. However, preliminary experiments indicate that free-form causal debate is inherently unstable when confronting commonsense traps. Standard debate frameworks not only fail to act as reliable reasoning enhancers but also frequently fall into a specific failure mode: *over-correction*. In this state, an agent that initially holds the correct answer is persuaded by its peer to adopt an incorrect one, effectively amplifying the trap rather than resolving it. This instability stems from a lack of structured constraints, which prevents critiquing agents from differentiating between core causal variables and misleading noise.

In this paper, we employ *Causal Variable Audit Debate* (CVA-Debate), a framework that transforms the free-form debate process into a structured causal variable audit. While prior work has shown that strong reasoning models can improve accuracy through dual-agent debate on formal causal benchmarks, we find that a local 1.5B-parameter model does not achieve comparable gains under our Chinese commonsense-trap setup. Therefore, we focus on why multi-agent causal debate fails in this regime and propose a structured variable audit to reduce over-correction. Our method improves over unstructured debate by requiring agents to explicitly disentangle the causal structure of a scenario before formulating a rebuttal. By identifying core causal variables (X), spurious variables (Z), and contextual reversal variables (C), the framework provides a logical anchor that prevents opinion drift and mitigates the backfire effects of over-reflection. We specifically emphasize the role of contextual reversal variables, which are often misidentified as noise by standard models but are essential for accurate causal inference.

To evaluate our approach, we introduce the *over-correction rate* (OCR) as a diagnostic metric to quantify the failure mechanisms of multi-agent reasoning. We conduct experiments on a newly constructed diagnostic set featuring Chinese commonsense causal traps. The results demonstrate

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that while free debate often collapses under the pressure of spurious cues, CVA-Debate provides a robust defense mechanism by forcing models to align their reasoning with a minimal sufficient evidence set.

The contributions of this work are as follows:

1. We propose CVA-Debate, a structured causal variable audit framework designed for resolving commonsense traps in natural language.
2. We introduce the concept of contextual reversal variables and the over-correction rate metric to analyze and diagnose the failure mechanisms of multi-agent causal debate.
3. We construct a Chinese Causal Trap Diagnostic Set and conduct categorical experiments to reveal the effective boundaries and robustness of the proposed method.

2. Related Work

In recent years, the paradigm of multi-agent debate has been widely applied to Large Language Models (LLMs). [Du et al. \(2024\)](#) demonstrated that a multi-agent debate framework—where multiple LLM instances iteratively generate, critique each other, and revise their answers—can eliminate uncertain hallucinated facts by reaching a consensus, thereby significantly enhancing the factuality and logical reasoning capabilities of the models. However, standard debate frameworks are often limited by the inherent biases and overconfidence of the models. To address this, the CounterFactual Multi-Agent Debate (CFMAD) framework of [Fang et al. \(2025\)](#) forces LLMs to adopt preset counterfactual stances, prompting them to generate abductive explanations for alternative answers and debate with a “critic,” effectively mitigating overconfidence and hallucination issues. Furthermore, recent studies have increasingly explored the integration of LLMs and causal inference. In the field of causal discovery, the MAC framework ([Le et al., 2024](#)) utilizes a multi-agent debate module to fuse structured data with contextual metadata, collaboratively selecting statistical algorithms and optimizing causal graphs. In the domain of formal causal inference, the recently proposed CRAWDAD ([Vamosi & Forkert, 2025](#)) demonstrates the effectiveness of dual-agent debate using advanced reasoning models (such as Qwen3-32B and DeepSeek-R1). Evaluated on the CLadder English dataset ([Jin et al., 2023](#)), which features rigorously formalized causal graphs, CRAWDAD proved that multi-agent debate can significantly improve models’ causal inference accuracy (especially on counterfactual queries).

Inspired by CRAWDAD, this study further investigates multi-agent debate by analyzing the causes of over-correction and attempting to provide a solution. Existing works

(such as CRAWDAD) mostly focus on improving accuracy through dual-agent debate on formal causal benchmarks like CLadder ([Jin et al., 2023](#)). In contrast, our research focuses on Chinese commonsense causal traps, exploring why multi-agent causal debate fails and why it produces over-correction due to variable identification errors.

Most prior works only report the accuracy improvements brought by multi-agent causal debates, but rarely analyze why debates can “backfire.” We found that in commonsense causal traps, the root cause of over-correction lies in “variable identification failure.” Unlike CRAWDAD, which relies on large 32B models to process formal yes/no questions, we utilize lightweight local models to tackle self-constructed Chinese natural language trap questions. In terms of evaluation metrics, in addition to accuracy, we introduce the correction rate and over-correction rate. To mitigate the backfire effect brought by debate, we propose a lightweight “Causal Variable Audit” framework, which explicitly disentangles and audits core variables, spurious variables, and contextual reversal variables prior to the debate.

3. Methodology

Conventional multi-agent debate frameworks allow models to freely generate answers and critique one another. Lacking structured constraints, the critiquing agent often fails to distinguish between “core cues” and “distracting cues.” It is prone to misidentifying outcome-determining key contexts (e.g., “Cut! That’s a wrap”) as noise and improperly discarding them. This leads to a specific failure mode: Over-Correction (i.e., altering an initially correct answer to an incorrect one). In contrast, our method transforms “free critique” into a “mandatory causal variable audit.” Prior to presenting a rebuttal, the model must explicitly disentangle the various variables within the question premise, clearly defining which variables the answer should be sensitive to and which it should ignore. This significantly mitigates the backfire effect of over-correction.

During the debate process, models are strictly required to explicitly decouple the cues embedded in natural language trap questions into the following structured dimensions:

- **X (Core Causal Variable):** The fundamental variable that genuinely determines the trajectory of the answer (i.e., if X is altered, the answer must inherently change).
- **Z (Spurious Variable):** The “superficial confounding variable” that appears highly relevant textually and easily induces intuitive bias in the model, but fundamentally does not dictate the final answer.
- **C (Contextual Reversal Variable):** This is the most critical addition proposed in our study to specifically

address over-correction. It refers to subtle cues (e.g., “however, in reality,” “cut, that’s a wrap”) that seemingly serve as background information but in fact completely reverse the trajectory of the entire scenario. Explicitly annotating C effectively prevents the model from mistakenly discarding crucial contextual reversals as noise.

- **MSE (Minimal Sufficient Evidence):** The minimal, core set of information genuinely required to answer the question after stripping away all confounding variables.

Centered around the aforementioned audit framework, this study designs a three-agent pipeline with clearly defined responsibilities:

- **Agent A (Intuitive Responder)** serves as the “target” and baseline. It utilizes minimalist prompts and responds rapidly based solely on the model’s pre-trained intuition, without engaging in deep deliberation. This maximizes the exposure of the original biases induced by spurious variables (Z) when Large Language Models confront commonsense causal traps.
- **Agent B (Causal Variable Auditor)** acts as the core of the methodology. Rather than hastily negating Agent A, it strictly adheres to the aforementioned framework to perform a structured extraction of X , Z , C , and MSE from the premise. Following the variable disentanglement, Agent B diagnoses Agent A’s misattributions (e.g., “Agent A erroneously relied on spurious variable Z while overlooking reversal variable C ”), and subsequently formulates a logically rigorous, audited recommended answer.
- **Agent C (Consistency Judge)** acts as the ultimate “gatekeeper” to prevent the blind acceptance of Agent B’s critiques. It is tasked with conducting a comprehensive consistency adjudication between A and B. Its core operating rule dictates that if Agent B mistakenly discards a “contextual reversal variable C ” by misclassifying it as a “spurious variable Z ,” Agent C must reject Agent B’s recommendation. The final verdict is output only when the concluding answer perfectly aligns with the MSE.

Figure 1 summarizes the information flow alongside the roles above.

Concurrently with the introduction of this methodology, the present study transcends the sole reliance on the single “Accuracy” metric by introducing the Over-Correction Rate. Through the interactive dynamic among Agents A, B, and C based on the $X/Z/C/MSE$ framework, this research can precisely capture whether the model exhibits

“successful correction,” “correct retention,” or catastrophic “over-correction” within Chinese causal traps. Consequently, this provides an interpretable diagnosis for the failure mechanisms underlying multi-agent reasoning.

4. Experiments

To verify the effectiveness of the proposed Causal Variable Audit Debate (CVA-Debate) in mitigating multi-agent over-correction, we designed targeted diagnostic experiments. The experiments focus on examining the model’s robustness against spurious variables and contextual reversals in natural language, as well as its internal debate dynamics.

4.1. Dataset Construction

Since existing causal reasoning benchmarks (e.g., CLadder (Jin et al., 2023)) primarily focus on Formal Causal Graphs and lack investigations into natural language contextual traps, this study constructs a specialized “Chinese Causal Trap Diagnostic Set.” This dataset comprises 20 meticulously designed representative commonsense trap scenarios. Rather than pursuing a large scale, it emphasizes the interpretable diagnosis of failure mechanisms.

Each data instance in the diagnostic set (see `final_20_traps.csv` in the released `final_data/` bundle) is manually annotated with fine granularity and contains the following core fields:

- **Question:** The natural language statement containing the trap.
- **Ground Truth:** The true causal outcome.
- **Structured Causal Variables:** Includes the core causal variable X , the spurious variable Z , and the contextual reversal variable C .
- **Expected Failure:** Records the intuitive bias most likely generated by large language models without intervention.

The dataset includes the following four types of traps that are highly prone to triggering “over-correction” in multi-agent systems:

1. **Spurious Correlation:** Induces errors by leveraging superficial identities or statistical co-occurrences.
2. **Emotion Word Misleading:** Emotional vocabulary in the premise conflicts with the true causal evaluation direction.
3. **Contextual Reversal:** A specific context later in the text instantly reverses the superficial event framework established earlier.

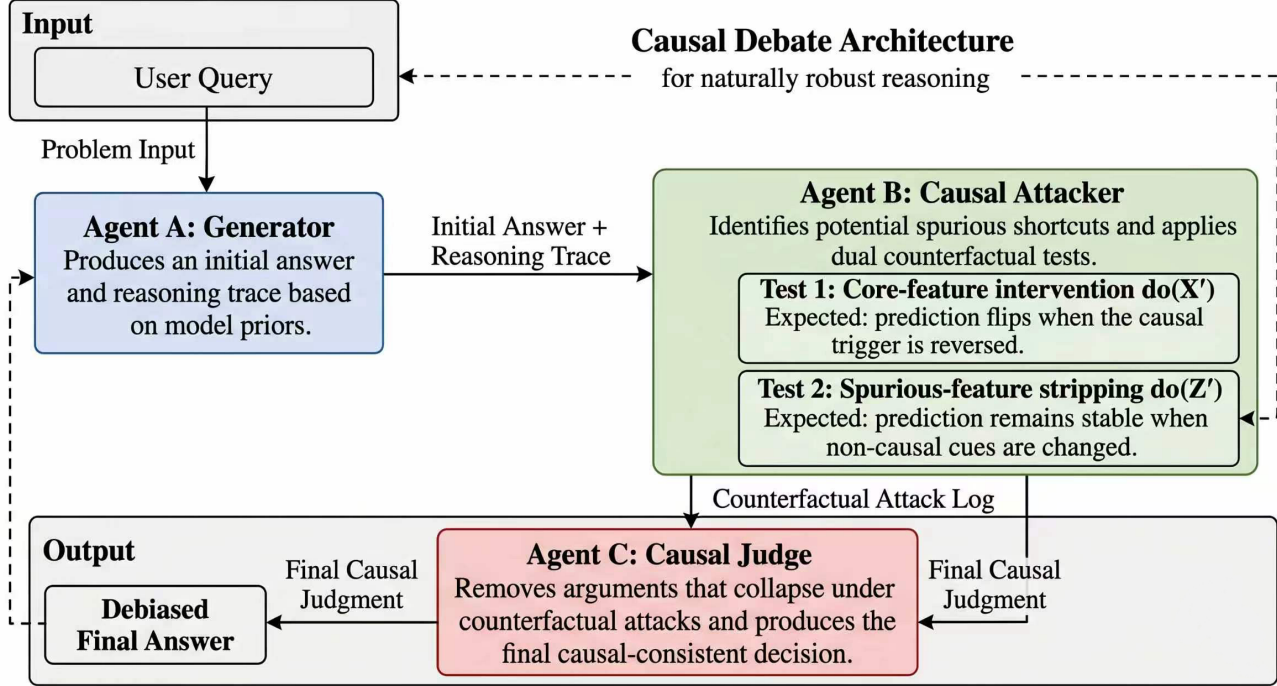


Figure 1. Causal debate architecture for CVA-Debate. Agent A (generator / intuitive responder) produces an initial answer and reasoning trace; Agent B (causal auditor) stress-tests reliance on core vs. spurious factors and records structured interventions consistent with X , Z , C , and MSE; Agent C (consistency judge) rejects updates that collapse under audit or mis-handle reversals, yielding a debaised final answer. Dashed arcs indicate optional multi-round refinement.

4. **Hard Causal Rules:** Relies on unshakeable objective reality rules, such as those grounded in physics, medicine, or physiology.

4.2. Baselines and Model Setup

To ensure a fair comparison and reveal the backfire mechanisms of multi-agent debate, this experiment sets up three reasoning paradigms:

1. **Vanilla (Zero-shot Reasoning):** Directly inputs the test question to the model and requests an answer and explanation. This baseline is utilized to expose the large model’s initial intuitive bias toward spurious variables derived from its pre-training data.
2. **Free Debate:** Employs a standard dual-agent causal debate process. Agents freely generate critiques and discuss with each other, without imposing any structured variable disentanglement constraints.
3. **CVA-Debate:** Introduces the structured causal variable audit process. Agent B is forced to extract $X/Z/C/MSE$ prior to critiquing, and Agent C makes the final anti-over-correction adjudication based on consistency rules.

Model Setup: For model selection, this experiment prioritizes the lightweight local open-source model Qwen2.5-1.5B, deployed for inference via the Ollama `/api/chat` interface. The rationale behind selecting a model with minimal parameters is twofold: first, it effectively controls the computational cost incurred by iterative multi-agent interactions; more importantly, small models possess weaker anti-interference capabilities against commonsense traps, enabling them to most authentically and acutely expose the micro-mechanisms of “over-correction” and “logical backfire.” This prevents large models (e.g., 72B or GPT-4) from brute-forcing traps via massive parameters, which would otherwise obscure the inherent flaws of the debate mechanism itself.

4.3. Evaluation Metrics

Distinct from the traditional evaluation paradigm that focuses solely on final accuracy, this study introduces a set of dynamic state metrics to diagnose the debate process:

- **Accuracy:** The proportion of questions for which the model outputs the correct final answer.
- **Correction Rate:** The proportion of samples that were initially answered incorrectly under the Vanilla mode but were successfully corrected after debate. This mea-

sures the positive gain of the debate.

- **Over-Correction Rate:** The proportion of samples that were initially answered correctly under the Vanilla mode but were led astray by the opponent’s rebuttal during debate, ultimately changing to an incorrect answer. This is the core metric for measuring the negative backfire effect of the debate system.

4.4. Human Evaluation Protocol

Because both the reasoning process and the final conclusions of causal traps are output in natural language, relying solely on automated script matching is highly prone to misjudgment. Therefore, this experiment adopts a rigorous human evaluation protocol.

For each test instance, human annotators record the states across the following dimensions:

1. **Binary Correctness:** Independently annotates the correctness of the final round output for the three methods, denoted as `vanilla.correct`, `free.debate.correct`, and `cva.debate.correct`, respectively.
2. **State Transition Tracking:** Based on the comparison between the baseline and the post-debate states, annotates whether the instance triggered a `correction.success` (0→1) or an `over.correction` (1→0).
3. **Fine-grained Failure Reason Attribution:** If the final output is incorrect (or if over-correction occurs), annotators must trace and classify the failure reasons from the following four dimensions:
 - **Variable Identification Error:** The model incorrectly treats a spurious variable Z as the core variable X .
 - **Contextual Reversal Misjudgment:** The model fails to capture the role of the contextual reversal variable C , potentially even discarding it as noise.
 - **Commonsense Deficit:** A break in the reasoning chain caused by the model’s lack of foundational world knowledge in physics or medicine.
 - **Contradiction:** The model’s reasoning process is logically sound, but the final conclusion drawn completely contradicts the explanation.

5. Experiment Results

5.1. Overall Accuracy and Performance Comparison

The results from three independent runs on the Chinese Commonsense Causal Trap Diagnostic Set are summarized in Table 1.

Table 1. Overall comparison of Free Debate versus CVA-Debate on the diagnostic set (aggregated over three runs).

Metric	Free Debate	CVA-Deb. (Ours)
Total Initial Correct	31	36
Total Final Correct	24	34
SCR	20.7% (6/29)	33.3% (8/24)
OCR	41.9% (13/31)	27.8% (10/36)
NDB	-7	-2

Experimental data show that, despite sharing the same base-model initial intuitions, traditional Free Debate is extremely unstable and yields a low ratio of final correct answers: SCR is only 20.7%, OCR reaches 41.9%, and NDB is -7, indicating severe regression. Thus nearly half of initially correct intuitions are sabotaged during debate. In contrast, CVA-Debate substantially improves SCR through structured auditing and reduces OCR by 14.1 percentage points (from 41.9% to 27.8%). With $NDB = -2$, the system moves toward stability and shows stronger causal defensibility: better performance preservation, stronger repair of initial errors, and interception of friendly fire from over-reflection, mitigating net loss and hardening the reasoning defense.

5.2. Analysis of Success and Failure Cases

Success Case 1 (Intercepting Spurious Correlation, State 11). In the “Anvil in a Zero-Gravity Chamber” experiment, Free Debate was misled by the textual description of the anvil’s weight (Z variable), inducing excessive skepticism. CVA, however, forced the identification of the zero-gravity physical environment (X variable), enabling Agent C to successfully shield the interference of weight and maintain the correct answer.

Success Case 2 (Capturing Contextual Reversal, State 01). In the “Entering a Bank Wearing a Mask” scenario, Agent B explicitly extracted the variable “The director shouted ‘Cut, wrap it up!’” (C variable). This successfully corrected the initial “robbery” intuition to the correct “filming a movie” conclusion.

Failure Case 1 (Divergent Backfire, State 10). In the “Broadcaster Dubbing on a Beach” question, the model over-interpreted the background information extracted by Agent B as a reversal variable (C). This caused the model to “overthink,” leading to an over-correction of an initially correct answer.

Failure Case 2 (Lexical Hijacking Leading to Audit Failure, State 00). In questions using high-polarity negative terms (e.g., “violently,” “greedily”) to describe positive reviews of a vacuum cleaner, the model’s attention mechanism was completely hijacked by the Z variables. Consequently, the audit report generated by Agent B was biased from its underlying logic, failing to form an effective defense.

6. Conclusion and Discussion

The core value of CVA-Debate lies in the establishment of a “logical anchor.” By mandating that Agent B list the Minimal Sufficient Evidence (MSE), the framework effectively prevents “opinion drift” during the debate process. The method achieves its optimal performance when the trap relies on a clear, single contextual reversal variable (C) that is semantically separable from spurious noise (Z).

Simultaneously, experiments reveal that multi-agent debate can degenerate into a “trap amplifier.” Within the CVA framework, backfiring primarily stems from the model’s “hyper-vigilance”: in its effort to complete the auditing task, the model tends to hallucinate a reversal variable C where no such reversal exists. This suggests that for LLMs, “analytical thinking,” when lacking the constraints of common sense, can result in higher-order hallucinations.

The use of small models presents inherent limitations. Experimental results with the 1.5B model expose the causal alignment boundaries of lightweight models. Although small models can adhere to the $X/Z/C$ auditing format, their underlying semantic representations sometimes fail to correctly distinguish between “physical reality” and “rhetorical devices.” This indicates that while a structured framework can guide reasoning, it cannot fully compensate for a deficiency in the base model’s world-model depth. Furthermore, in the Chinese-context commonsense-trap setting of this work, unstructured Free Debate already fails to meet stability requirements (high over-correction and net loss); the introduction of CVA-Debate effectively targets this gap.

This research clarifies the boundary between “semantic causal prompting” and “formal causal inference.” CVA-Debate is a training-free heuristic method that simulates behaviors associated with Pearl’s ladder of causation; however, its essence remains rooted in linguistic probability. When the “variable identification” stage fails, the entire logical pipeline collapses. Consequently, it should be viewed as an improvement scheme for causal robustness rather than an absolute tool for formal proof.

In summary, this study shows that multi-agent counterfactual debate is not an inherently reliable enhancer for causal reasoning: its effectiveness depends heavily on whether the system can accurately identify and decouple key causal variables in natural language.

Experimental results show that traditional “Free Debate” exhibits significant vulnerability when facing Chinese commonsense traps, with its Over-Correction Rate (OCR) reaching as high as 41.9%, making it highly susceptible to logical backfire. In contrast, our proposed CVA-Debate provides an interpretable, lightweight, and training-free framework for improving LLM causal robustness by explicitly distinguishing among Core Variables (X), Spurious Variables

(Z), and Contextual Reversal Variables (C). This framework reduces the “mis-killing” of correct answers within the debate system and enhances the stability of causal reasoning by establishing logical anchors.

Accessibility

We structure the main quantitative results as tables with textual headers so they remain interpretable when reflowed or read aloud. For any subsequent camera-ready figures (e.g., pipeline diagrams), we will follow ICML guidance on colorblind-safe palettes, sufficient contrast, and descriptive captions; we refer readers to <https://icml.cc/> for conference-specific accessibility recommendations.

Software and Data

To support reproducibility and external audit, we release experimental artifacts under the `final_data/` directory at the root of our supplementary archive (paths relative to that root). The Chinese Causal Trap diagnostic items are in `final_20_traps.csv`. Raw multi-run debate transcripts and numeric summaries are provided as `cva_debate_results-{0,1,2}.csv`, `free_debate_results-{0,1,2}.csv`, and the aggregated sheet `sum.csv`. The driver script `FINAL.py` documents how local inference (e.g., via Ollama) was orchestrated for batch evaluation; `README.md` and two supplementary Chinese-language Markdown reports in `final_data/` (study protocol and experiment-analysis write-ups; filenames begin with `CVA_Debate_`) further summarize design and analysis. For the anonymized submission, any public URL or persistent deposit should omit identifying affiliation; after acceptance, we will register a stable repository or archive link aligned with camera-ready requirements.

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Impact Statement

This work studies when multi-agent deliberation improves or harms causal judgments in language models, with emphasis on *over-correction* and variable misidentification. Improved debate protocols could make automated reasoning support more reliable in education, legal assistance, or decision-support settings; conversely, a finer understanding of persuasion dynamics in multi-agent LLM systems could inform safeguards against coordinated misinformation or

manipulation, including in culturally specific (e.g., Chinese) commonsense domains where spurious cues differ from English-centric benchmarks. Our diagnostic set is small and intended for analysis rather than training high-capacity models; we nevertheless caution against using debater-style pipelines in high-stakes settings without human oversight and outcome monitoring. We see net benefit in making failure modes (including debate-induced backfire) measurable and interpretable, so practitioners can calibrate when structured auditing helps versus when it amplifies errors.

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A. Released artifacts (`final_data/`)

The following sections summarize the released artifacts, the schema of the Chinese Causal Trap diagnostic set, the high-level CVA-Debate control flow, and formal definitions of the dynamic metrics used in the main text. All path names are ASCII so that the manuscript compiles under $\text{\LaTeX}$; the actual item text in the released CSV is in Chinese.

We distribute the following files relative to the `final_data/` root:

- `final_20_traps.csv`: one row per trap item, with fields described in Section B.
- `cva_debate_results- $\{0, 1, 2\}$ .csv` and `free_debate_results- $\{0, 1, 2\}$ .csv`: raw logs for three independent runs (indices 0–2).
- `sum.csv`: run-level aggregates used to report OCR, SCR, and related counters in Table 1 of the main paper.
- `FINAL.py`: batch driver for local inference (e.g., `Ollama /api/chat`); hyperparameters such as temperature should be matched to the submission revision if results are recomputed.
- `README.md`: short index of the above; additional Markdown notes with the prefix `CVA_Debate_` document the study protocol and quantitative analysis in Chinese.

B. Diagnostic item schema

Each CSV row contains at least the following conceptual columns (exact header spelling follows the released export; some builds use English field labels in documentation while the spreadsheet content remains Chinese):

- **Question**: full natural-language premise containing the trap.
- **Best answer / ground truth**: the adjudicated correct outcome for the task (e.g., sentiment, event label, yes/no).
- **Trap category (spurious pattern)**: a short tag for the failure mode the item targets (aligned with the four families in Section 4.1).
- **Z (spurious variables)**: surface cues that encourage the wrong shortcut.
- **C (contextual reversal)**: late-appearing clause that overrides the superficial frame (may be empty on some items).
- **X (core causal variables)**: facts that should actually determine the answer once confounders are stripped.

C. Trap taxonomy (illustrative mapping)

We group items so that human annotators and automatic bookkeeping share the same labels:

1. **Spurious correlation**: high textual co-occurrence of a stereotype and the wrong inference (e.g., elite attire vs. actual behavior).
2. **Emotion-word misleading**: polarized vocabulary conflicts with the factual valence of the outcome.
3. **Contextual reversal**: a key clause negates the scenario suggested by the opening (e.g., “cut—wrap” on a film set).
4. **Hard causal rules**: physics, physiology, or other domain laws that dominate salient but irrelevant biographical detail.

D. CVA-Debate control flow

Let q denote the user question. The three-agent pipeline executes in order:

1. **Agent A (intuition)**: produce an answer a_0 and a short rationale under a minimal prompt (no audit template).
2. **Agent B (audit)**: extract (X, Z, C, MSE) from q , diagnose whether a_0 relied on spurious Z or missed reversal C , and output a revised answer a_B together with the structured audit trace.

3. **Agent C (consistency):** verify that a_B respects every reversal in C and is supported by MSE; if B misclassifies C as Z , reject a_B and fall back to a_0 or request a repair round (implementation detail fixed per run sheet).
4. The *final* system output is the last message authorized by Agent C after the above checks.

Free Debate omits steps that require explicit (X, Z, C, MSE) extraction before substantive critique; Vanilla omits multi-agent critique entirely.

E. Metric definitions

Let $\mathcal{I}$ index items in the diagnostic set. For each method $M \in \{\text{Vanilla}, \text{Free}, \text{CVA}\}$ and item i , let $c_i^{(M)} \in \{0, 1\}$ indicate whether the *final* model output is correct under M , and let $c_i^{(0)}$ denote Vanilla correctness (initial intuition).

$$\begin{aligned} \text{SCR}_M &= \frac{\sum_{i \in \mathcal{I}} \mathbf{1}[c_i^{(0)} = 0 \wedge c_i^{(M)} = 1]}{\sum_{i \in \mathcal{I}} \mathbf{1}[c_i^{(0)} = 0]}, \\ \text{OCR}_M &= \frac{\sum_{i \in \mathcal{I}} \mathbf{1}[c_i^{(0)} = 1 \wedge c_i^{(M)} = 0]}{\sum_{i \in \mathcal{I}} \mathbf{1}[c_i^{(0)} = 1]}, \\ \text{NDB}_M &= \sum_{i \in \mathcal{I}} c_i^{(M)} - \sum_{i \in \mathcal{I}} c_i^{(0)} \quad (\text{aggregate change in correct count}). \end{aligned}$$

The table in the main paper reports these quantities after merging three random seeds; per-run CSVs allow recomputing dispersion across runs if needed.

F. Human evaluation fields

Annotators store binary correctness flags (e.g., `vanilla_correct`, `free_debate_correct`, `cva_debate_correct`), transition codes `correction_success` (0→1) and `over_correction` (1→0), and, when applicable, failure tags for variable mix-up, missed reversal, commonsense gaps, or internal contradiction, as described in Section 4.4.